

12th August, 2024

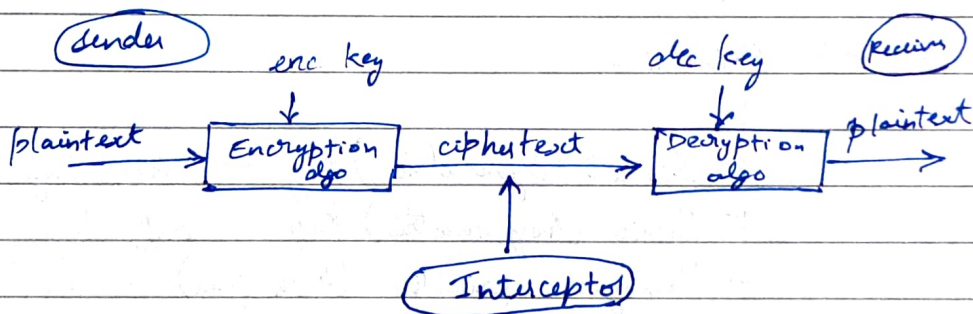


Applied Cryptography

The discipline that embodies the principles, means and methods for the transformation of data in order to hide their semantic content, prevent their unauthorized use, or prevent their undetected modification.

Applications : Encryption/Decryption in email
Electronic money
Encryption in whatsapp
sim card authentication

→ Crypto System



→ Services

Confidentiality : info kept priv & secure

Integrity : data not modified, deleted or added

Availability : systems avail. to whom requires them

Authenticity : Provides verification of identities

X Accountability : assurance

Non repudiation : Person that sent msg cannot
refuse doing so.

→ Protocols

* used for app-level data transport security

- * - key establishment
- Entity authentication
- sym encryp & msg auth
- material const
- Non-repudiation methods
- Secure sharing methods
- Secure multi party computation

eg: IPsec, TLS/SSL, SSH

→ Primitive

- Secret (Symmetric) key - same key
- Public (Asymmetric) key - diff key
- Unkeyed primitive (Hash func) - no key

→ Evaluation

- Level of security
- Functionality
- Mode of operation
- Performance
- Ease of implementation

15th August, 2024

Caesar Cipher

q. Shift 3
encrypt msg: "We hide at noon"

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

We hide at noon

ZH U~~X~~ GH DW QRRQ

q. Decrypt GZD KNK YDX MFW JXA
shift 5

ans: BUY FIF ~~T~~YS HAR ESA

q. BEGIN THE ATTACK NOW
shift 3 EHJLQ WKH PWDFN QRZ

Polyalphabetic Substitution cipher

Playfair cipher
Vignère cipher

→ Vignère cipher

eg: Plain text : I attack
key : 234

1 A T T A C K
2 3 4 2 3 4 2
cipher K D X V D G M

eg: Plain text : I attack
key : EXAM

Plain text I A T T A C K
Plain value 8 0 19 19 0 2 10
Key E X A M E X A
key value 4 23 0 12 4 23 0

cipher value 12 23 19 5 4 25 10
cipher M X T F E Z K

9.

Plain text: A L L I S W E L L
Plain val: 0 11 11 8 18 22 4 11 11
key: C A K E C A K E C
key val: 2 0 10 4 2 0 10 4 2

cipher val: 2 11 21 12 20 22 14 15 13
cipher C L V M U W O P N

A B C D E F G H I J K L M N O P Q R S T
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

U V W X Y Z
20 21 22 23 24 25

9.

txt	s	h	e		i	s		l	i	s	t	e	n	i	n	g
txt val	18	7	4		8	18		11	8	18	19	4	13	8	6	9
key	p	a	s		c	a		p	a	s		c	a	p	a	
key val	15	0	18		2	0		11	15	0	18	2	0	11	15	0
cipher val:	7	7	22		10	18		22	23	18	11	6	13	19	2	6
cipher	q	q	w		k	s		w	x	s		l	g	n	t	c

Play Fair Cipher

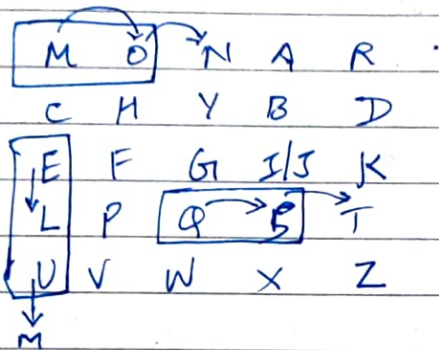
eg: Plain text : attack
 diagram : at ta ck
 key: monarchy

M	O	N	A	R
C	H	X	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

at ta ck
 RS SR DE

g. Plain text : MOSQUE
key : MONARCHY

MO SQ UE
ON TS ML



Repeating $\Rightarrow$ Filler (X)
same col $\Rightarrow$ $\downarrow$ wrap
same row $\Rightarrow$ $\rightarrow$ wrap
rectangle $\Rightarrow$ $\rightleftarrows$ wrap

g. TE LL HI MA BO UT ME

~~TE LX XL HI MA BO UT~~

TE LX LH IM AB OV TM EX

key : ENCRYPT

E N C R Y T
P T A B D
F G H I J K
L M O P Q
R S V W X Z

TE LX LH IM AB OV TM EX
PN QU OF GQ BD LW GV RU

q. Plain text : instruments
key : Monarchy

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

cipher

IN	ST	RU	ME	NT	SX
GA	TL	MZ	CL	RQ	XA

Transposition cipher

- Rail Fence
- Block (Single Columnar)
- Double Columnar

→ Rail Fence

eg: Plain text : "this is a secret" message"
key = 2

T	I	I	A	E	R	T	E	S	G
H	S	S	S	C	E	M	S	A	E

TIIAERTESGHSSCEMSAE

eg: Plain text: "defend the east wall"
key: 3

D	N	E	T	L					
E	E	D	H	E	S	W	L	X	
F	T		N		A			X	

DNETLEEDHESWLXFTAAX

eg: cipher: T E K O O H R A C I R M N R E A T A N F T E T Y T
key: 4 G H H

T	E	K		O		D		
H	R	A	C	I	R	M	N	R
E	A	T	A	N	E	I	E	T
Y		T		G		H		H

Plain: They are attacking from the north.

→ Columnar Transposition

eg: message: WE ARE DISCOVERED. FLEE AT ONCE
keyword: ZEBRAS

6	3	2	4	1	5
<u>Z</u>	<u>E</u>	<u>B</u>	<u>R</u>	<u>A</u>	<u>S</u>
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	-	-	-	-	-

cipher: EVLN_ACDT_ESEA_R0F0_DEEC_WIRGG

eg: Plain: GEEKS FOR GEEKS
key: BOAT

	2	3	1	4
	B	O	A	T
G	E	E	K	
S	F	O	R	
G	E	E	K	
S	_	_	_	

cipher: ~~E~~O~~E~~ _ ~~B~~G~~S~~G~~S~~ ~~E~~F~~E~~ _ ~~K~~R~~K~~ _

(28)
decrypt: TRLEELIGCIGEHALANTNCTECYENEN
key: ANALYST (7)

$$7 \times 4 = 28$$

	A	A	L	N	S	T	Y
-	T	E	C	H	N	T	E
-	R	L	I	A	T	E	N
-	L	I	G	L	N	C	E
-	E	G	E	N	C	Y	N

	A	N	A	L	Y	S	T
T	H	E	C	E	N	T	
R	A	L	I	N	T	E	
L	L	I	G	E	N	C	
E	A	G	E	N	C	Y	

THE CENTRAL INTELLIGENCE AGENCY

→ Double Transposition cipher

Plain: We are discovered flee at once

key 1: zebras

key 2: help

for key 1:

6	3	2	4	1	5
Z	E	B	R	A	S
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	Q	K	I	E	U

cipher 1: EVLNEACDTKESEAQR OFDJ DEECUWIR EE

for key 2:

2	1	3	4
H	E	L	P
E	V	L	N
E	A	C	D
T	K	E	S
E	A	Q	R
O	F	O	J
D	E	E	C
U	W	I	R
E	E	-	-

cipher 2: VAKAFEWEEET EORVELCEQOEI
ADSRJCR

5th September, 2024

Basic Crypto Primitives $\rightarrow \begin{cases} \text{Secret key} \\ \text{(same key)} \end{cases}$

I
 $\rightarrow$ Symmetric key : * single key b/w sender & receiver
* used for privacy & confidentiality
* " " both in-transit & at rest
* key needs to be stored securely
* secure channel req to transfer key

Design Principles : i) Block size
ii) Key size
iii) no. of rounds
iv) subkey generation
v) Round func.
vi) fast software en/decryption

eg: Block ciphers (AES, DES); stream ciphers (RC4)

Caesar cipher
PRNG

$\rightarrow$ Public (Asymmetric): each participant $\begin{cases} \text{priv key} \\ \text{public key} \end{cases}$

enc - from public key, decrypted using priv key
enc from priv key, decrypted using public key

ingredients : Plain Text
Ency Algo
Priv Key
Pub Key
Decy Algo
Cipher Text

- * Receiver creates the pair of keys
- * method starts with receiver
- * Public key: public to everyone
Private key: only known to receiver
- * To verify sender — Digital sign.
- * use it to exchange symmetric key

eg: Diffie - Hellman

ECC

RSA

El Gamal

PKI: Public Key Infrastructure

→ Unkeyed Primitives (Hash funcs)

- Hash funcs:
- * one-way cryptography
 - * irreversibly "encrypt" info
 - * gives digital finger print
 - * used for message integrity
 - * one way permutations
 - * No key, plain text is not recoverable

used for: Passwords, certificates

eg: MD5, SHA1, SHA2, Whirlpool, Blake etc

→ Math

Keyless: Bit-swapping

Shared Key: around binary Galois Field

$GF(2^k)$, permutation, matrix multiplication

Public key: number theory

→ Cryptographic Primitive Evaluation

Level of security
Functionality
Mode of operation
Performance
Ease of implementation

L4 # Crypto analysis: study of cryptographic algo
or breaking of those secret codes

Person practicing → Cryptoanalyst

→ Exhaustive Key Search : Try all possible keys
or test each to see if it's correct



Prevention : large key space

→ Secure : A cryptosystem is secure if the
best known attack is to try all possible keys

insecure ⇒ if any shortcut attack is known.

By def: insecure might be harder to break
than secure.

→ Theoretical cryptanalysis

Suppose cipher has 100 bit key

⇒ Key space size : 2^{100}

on avg exhaustive key search } $\frac{2^{100}}{2} = 2^{99}$ keys
hacker tests

Cryptoanalysis : cryptanalytic attackers rely on nature of algo + some knowledge of general characteristics of plain text

Brute Force attack : Attacker tries every possible key on a piece of cipher key text

— Attacks on Caesar Cipher

- * Letter freq analysis (same plaintext maps to same cipher text)
- * Shift cipher is not secure

— Cryptanalysis of Vigenere Cipher

- * Shift cipher involves replacing each letter in the message by a fixed no. of positions further along the alphabet

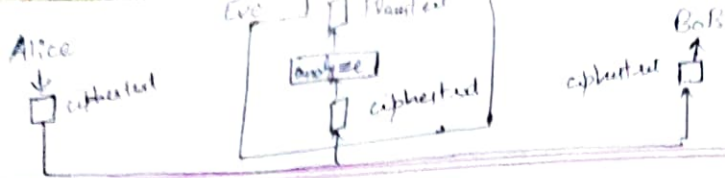
Vigenere masks the freq. with which a char appears in a lang: one letter in cipher corresponds to multiple letters in plain text

Makes "freq. analysis" more difficult

Any msg encrypted, collection of many shift ciphers as letters in the key.

Gryptoanalysis

┌──────────┴──────────┬──────────┬──────────┬──────────┐
Ciphertext only Known plaintext Chosen plaintext Chosen cipher text

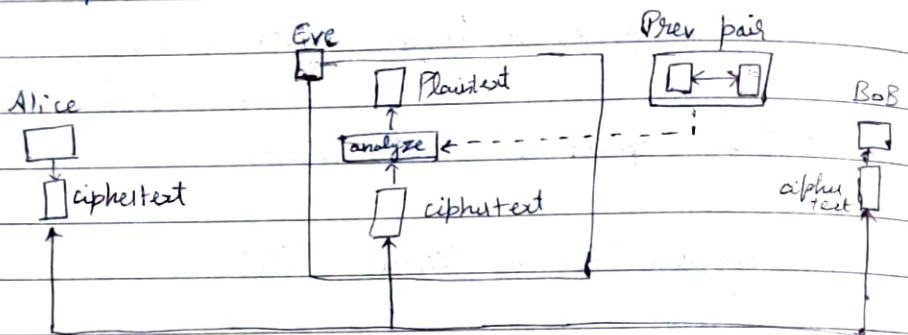


① Ciphertext - Only Attack

- * known: ciphertext, algorithm
- * statistical attack (freq analysis)
- * need to identify plain text

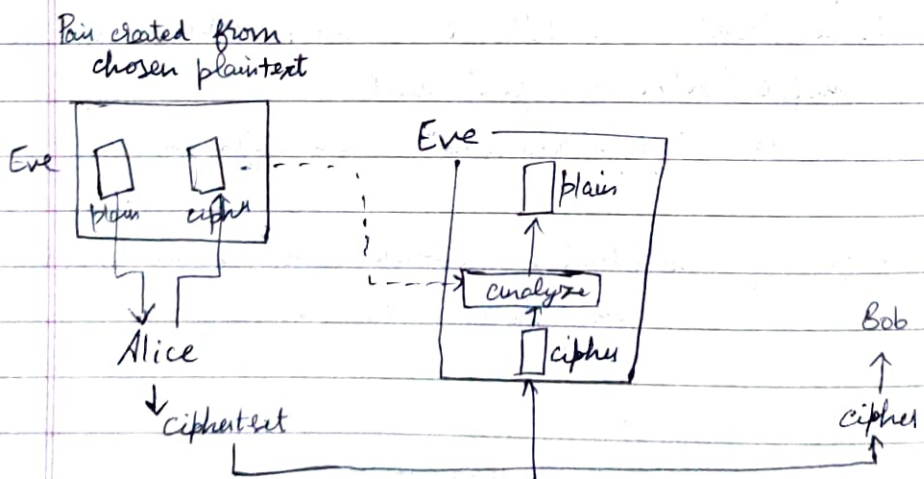
② Known - Plaintext Attack

- * known: plaintext, ciphertext (of suspect)



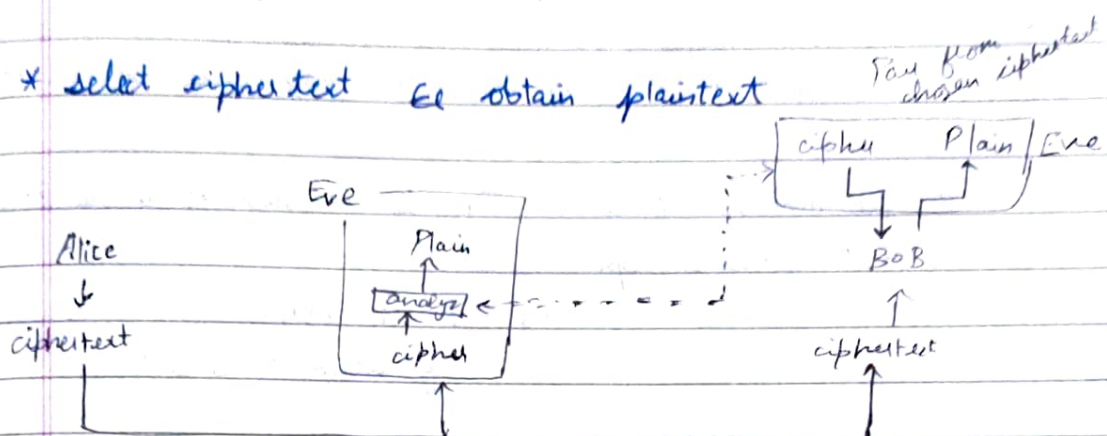
③ Chosen - Plaintext attack

- * select plaintext & obtain ciphertext



④ Chosen - ciphertext attacks

* select ciphertext & obtain plaintext



15 # Math for Cryptography

→ Integer Division

$$a \div n$$

$$a = q \times n + r$$

→ Modular Arithmetic

System of arithmetic for integers, where values reset to 0 and begin to increase again after reaching certain predefined value called modulus.

* Two nos. equivalent mod n if their difference is multiple of n

eg: 7 and 10 are equivalent mod 3

$$7 \div 3 \equiv 10 \div 3$$

→ Set of Residues: The modulo operation creates a set which in modular arithmetic is referred to as set of least residues modulo n , Z_n

$$Z_n = \{0, 1, 2, 3, \dots, (n-1)\}$$

→ Modular Inversion

↓
Integers in modular arithmetic may have inverse nos.

* The inverse of x in $\mathbb{Z}_N$ is a number y in $\mathbb{Z}_N$, that satisfies eqn.

$$x \cdot y = 1 \quad (\text{in } \mathbb{Z}_N)$$

y denoted as x^{-1}

* x is invertible in $\mathbb{Z}_N$ iff x & N are relatively prime

→ Calculating inverse numbers in $\mathbb{Z}_N$

① Naive method (brute force, simple but slow)

* trying all nos. from $\{0, \dots, m-1\}$

for every x in the set calculate $a \times x \bmod m$

* modular multiplicative inverse of modulo m is value of x for which remainder is $= 1$.

② Extended Euclidean Algo (faster)

Bézout's identity : assume a & b integers.

$$\exists x, y \mid \boxed{ax + by = \gcd(a, b)}$$

Euclidean

- * $\gcd(a, 0) = a$
- * $\gcd(a, b) = \gcd(b, r)$; r is remainder of a/b
- * if $\gcd(a, b) = 1$, a, b are relatively prime

→ Extended Euclidean Algo

$$s \times a + t \times b = \gcd(a, b)$$

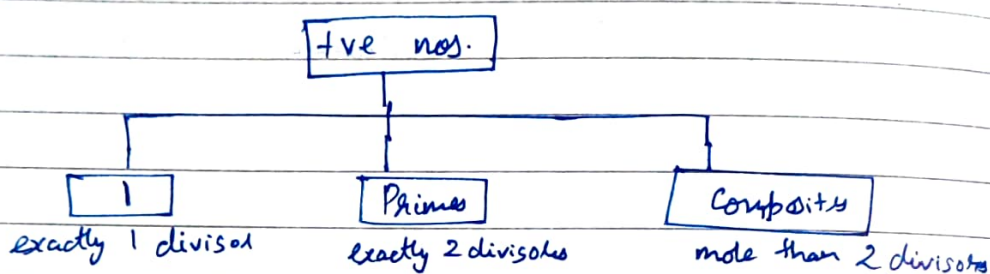
can calculate $\gcd(a, b)$ & at the same time
calculate s & t

→ Prime Numbers

* Asymmetric-key cryptography uses primes extensively

Prime nos. significance:

- i) implementation improvements
- ii) primality certificates
- iii) better versions of basic algos to speed up implementations.



→ Checking Primeness

if no. is divisible by all primes $< \sqrt{n} \Rightarrow$ Not Prime

→ Euler's Phi-Function (Euler's totient func) $\phi(n)$

1. $\phi(1) = 0$
2. $\phi(p) = p - 1$ if p is prime
3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m & n are relatively prime
4. $\phi(p^e) = p^e - p^{e-1}$ if p is prime

difficulty of finding $\phi(n)$ depends on difficulty of finding factorization of n .

6th September, 2024

→ Fermat's little theorem

If p is a prime no., then $a^{(p-1)} \equiv 1 \pmod{p}$

Applications : i) Primality Testing

ii) Cryptography

iii) Random number generation

Probability : likelihood of occurrence of an event

↳ ratio of no. of favourable outcomes to total no. of outcomes of an event

→ Conditional Probability : Prob. of an event occurring given that another event has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

joint prob. of events A & B

cond. prob. of
event A occurring given
that B has occurred

Prob of event B

→ Probability in cryptography

- i) Random Number Generator
- ii) Probabilistic Encryption
- iii) Error Detection
- iv) Security Proofs
- v) Hash functions
- vi) Zero-Knowledge Proofs
- vii) Statistical Tests

→ Modern Cryptography

* Design, analysis and implementation of mathematical techniques for securing info, systems & distributed computations against adversarial attacks.

- Scope: Data Integrity, authentication, protocol
Public key setting
Group Communication
More Complicated trust models.

- Applications: i) Password based authentication, password hashing
ii) Secure credit card transn. over the internet
iii) encrypted wifi
iv) disk encryption
v) digitally signed sw updates
vi) Bitcoin.

- Definition Of Cryptosystem

An encrypted system (cipher) is a 5 tuple (P, C, K, E, D)

P : finite set of possible plaintexts

C : finite set of possible ciphertexts

K : finite set of possible keys

for each $k \in K$ there is an encryption rule
 $e_k \in E$ and a corresponding decryption rule
 $d_k \in D$.

Each $e_k: P \rightarrow C$ and

$d_k: C \rightarrow P$ are funcs such that
 $d_k(e_k(x)) = x$ for every $x \in P$

→ Kerckhoffs' Principle

"A cryptographic system should be secure even if everything about the system, except the key is public knowledge."

→ An Ideal cryptographic system

- * Is a single all-purpose system which is practical for use from highest HQ to individual soldier
- * It is secure no matter how much message traffic is sent using the system. It is easy to use without special training
- * It presents no logistics problems in keeping the users supplied with the system's keys.
- * It operates under all weather conditions, on all means of comm. and in dark. Little of value is compromised if enemy captures the system
- * No System exists that meets all these requirements

→ Perfectly Secure Encryption

- * An encryption scheme (Gen, Enc, Dec) over a message space M is perfectly secure if for every probability distribution over M , every message $m \in M$, and every ciphertext $c \in C$ for which

$$Pr[C=c] > 0:$$

$$Pr[M=m | C=c] = Pr[M=m]$$

→ Basic Principles of Modern Cryptography

- Principle 1** The first step in solving any cryptographic problem is the formulation of a rigorous & precise definition of security.
- Principle 2** When security of a cryptographic construction relies on an unproven assumption, this assumption must be precisely stated. Furthermore the assumption should be as minimal as possible.
- Principle 3** Cryptographic constructions should be accompanied by a rigorous proof of security w.r.t a definition formulated according to principle 1 & relative to an assumption stated in principle 2.

September 7, 2021

L7 Perfect Secrecy

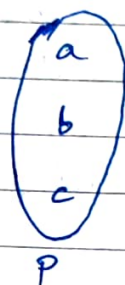
Encryption

- * For a given key, encryption (E_k) defines an injective mapping b/w plaintext set (P) & ciphertext set (C)
- * We assume that the key & plaintext are independent
- * Alice picks a plaintext $x \in P$ & encrypts it to obtain ciphertext $y \in C$

→ Plain Text Distribution

* Let X be a discrete random variable over the set P

Let $P_X[X=x]$ be the probability that x is chosen
This prob. may depend on language



$$P_X[a] + P_X[b] + P_X[c] = 1$$

→ Key Distribution

→ CipherText Distribution

* Let Y be a discrete random var over the set C

* The prob. of obtaining a particular ciphertext y depends on the plaintext & key probabilities

$$P_X[Y=y] = \sum_k P_X(k) P_X(d_k(y))$$

$$P_X[Y=P] + P_X[Y=Q] + P_X[Y=R] = 1$$

→ Attackers Probabilities

↓
want to determine plaintext x

Case 1: Attacker doesn't have y (a priori Probability)

- Prob of determining x is simply $P_X[x]$
- Depends on Plaintext distribution (eg. lang characteristics)

Case 2: Attacker has y (a posteriori probability)

- Prob of determining x is simply $P_X[x|y]$

→ Posteriori Probabilities $P_X[X=x | Y=y]$ - Bayes' Theorem

$$P_X[x|y] = \frac{P_X[x] \times P_Y[y|x]}{P_Y[y]}$$

prob. of plaintext prob. of ciphertext prob. that y is obtained given x depends on the keys which provide such a mapping

$$P_Y[y|x] = \sum_{\{k: d_k(y) = x\}} P_X[k]$$

→ Definitions:

* An encryption scheme is defined by 3 algo: Gen, Enc, Dec as well as a specification of a message space M with $|M| > 1$.

* The key-generation algorithm Gen is a probabilistic algorithm that outputs a key k chosen according to some distribution.

- * The encryption algo Enc takes as input a key $k \in K$ and a message $m \in M$ & outputs a ciphertext c : $\text{Enc}_k(m)$
- * The decryption algo Dec takes as input a key $k \in K$ and a ciphertext $c \in C$ and outputs a message $m \in M$, we denote this by $\text{Dec}_k(c)$

Perfect Secrecy $\Rightarrow$ Size of Message Space =
 Size of Key Space =
 Size of ciphertext space

Every msg is equally likely

achieved when

$$\text{Posteriori Prob} = \text{Priori Prob}$$

$$P_A[x|y] = P_A[x]$$

i.e attacker learns nothing from ciphertext.

$\Rightarrow$ Goal of Secure Encryption

- * An encryption scheme is secure if it is:

- impossible for the attacker to learn the key
- impossible for attacker to learn plaintext from ciphertext
- impossible for attacker to learn any character of the plaintext from ciphertext

- * Regardless of any prior info the attacker has about plaintext, the ciphertext should leak no additional info about plaintext

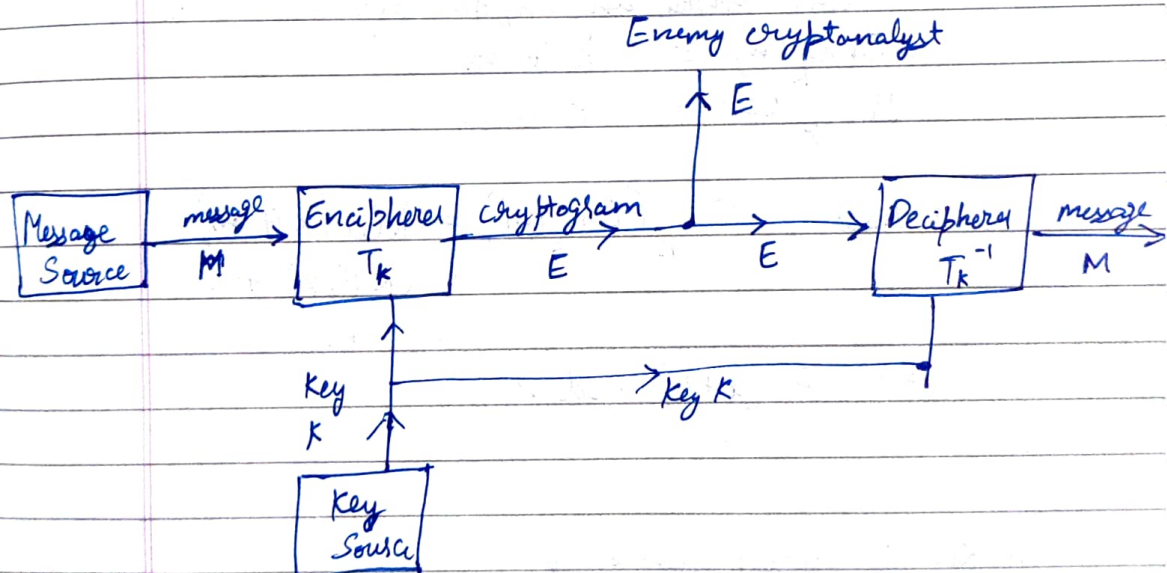
→ Perfect Secrecy Definition 2

An enc scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ over message-space M and key space K is said to be perfectly secret if for all messages $m_1, m_2 \in M$ & all ciphertexts c we have

$$\Pr[k \leftarrow \text{Gen} : \text{Enc}_k(m_1) = c] = \Pr[k \leftarrow \text{Gen} : \text{Enc}_k(m_2) = c]$$

→ attacker cannot even get any partial info about the message from the ciphertext since all messages give identical distributions on the ciphertext.

→ Schematic of general secrecy system



→ Shannon's Theorem: Suppose (P, C, K, E, D) is a cryptosystem, where $|K| = |C| = |P|$. Then the cryptosystem provides perfect secrecy iff

- every key is used with equal prob $1/|K|$ &
- for every $x \in P$ & $y \in C$ there is a unique key such that $e_k(x) = y$.

L8

OTP : One Time Pad

1

provably secure cryptosystem
by Shannon's Theorem

msg is represented as a binary string

- * Basic idea: Extend Vigenere cipher,
use a key as long as plaintext
- * No repeat, cannot be broken by finding
key len + freq analysis
- * Encryption: XOR b/w plaintext and key
- * Decryption: XOR b/w ciphertext and key

Let $M = \{0, 1\}^n$

Gen: choose an uniform key $k \in \{0, 1\}^n$

$$\text{Enc}_k(m) = k \oplus m$$

$$\text{Dec}_k(c) = k \oplus c$$

$$\begin{aligned}\text{Correctness : } \text{Dec}_k(\text{Enc}_k(m)) &= k \oplus (k \oplus m) \\ &= (k \oplus k) \oplus m \\ &= m\end{aligned}$$

$$(a \oplus b) \equiv (a + b) \bmod 2$$

- * Key is chosen randomly
- * Plaintext $X = (x_1, x_2, \dots, x_n)$
- * Key $K = (k_1, k_2, \dots, k_n)$
- * ciphertext $Y = (y_1, y_2, \dots, y_n)$

$$\begin{aligned}e_k(X) &= (x_1 + k_1, x_2 + k_2, \dots, x_n + k_n) \bmod m \\ d_k(Y) &= (y_1 - k_1, y_2 - k_2, \dots, y_n - k_n) \bmod m\end{aligned}$$

a	b	$c = a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

→ OTP Rules

- * OTP should consist of truly random nos.
- * Precisely 2 copies of OTP should exist
- * The OTP should only be used once
- * Both copies of OTP should be destroyed immediately after use

→ OTP is unbreakable

- * key is atleast as long as the message
- * key is truly random (not auto generated)
- * each key should only be used once & destroyed by sender & receiver.
- * There should only be 2 copies of the key (1 for sender, 1 for receiver).

→ Key Randomness in OTP

What if key is not chosen randomly

- it is not OTP anymore
- does not have perfect secrecy
- can be broken

- * The key in OTP should never be reused - if it is reused, it is Two Time Pad & insecure.